

From pixels to *the full picture*.

Satellite analytics, large geospatial datasets and machine learning for deforestation, crops and carbon, mapped with a cartographer's eye.

ABOUT

From paper maps and cadastre to satellite time series and foundation models.

I grew up in Porto and now live in Sofia. I've drawn cartographic and cadastral maps, assessed environmental impacts, and spent the last several years building crop analytics, learning how crops behave across every continent and season.

Today I help organisations turn satellite imagery, open and commercial, into evidence they can act on.

HOW I CAN HELP

Environmental due diligence

Deforestation screening against the EUDR cutoff, protected areas, soil and climate exposure.

Carbon & soil MRV

Practice evidence such as green cover between crops; persistent land-use change.

Crop & field monitoring

Crop type, phenology, anomalies and declaration checks, from one farm to a country.

Cartography

Maps designed to be understood, for reports, publications and print.

PATH

- **Cartography & cadastre**
where every boundary had to be right
- **Environmental assessment**
impact studies and baselines
- **Agtech**
several years of crop analytics at scale
- **Independent**
satellite analytics for environment & agriculture

WORKED WITH

Multinational food, beverage and retail companies, agricultural and forestry producers in Brazil, agricultural insurers, and the World Health Organization.

TOOLS

Python (xarray, dask, geopandas, scikit-learn, PyTorch) · QGIS · Google Earth Engine · STAC / COG

DATA

Sentinel-1/2/3/5P · Landsat · Maxar · Planet · LPIS · Copernicus DEM

LANGUAGES

English · Portuguese · Bulgarian · Spanish

BASED IN

Sofia, Bulgaria

Have land you need to understand?

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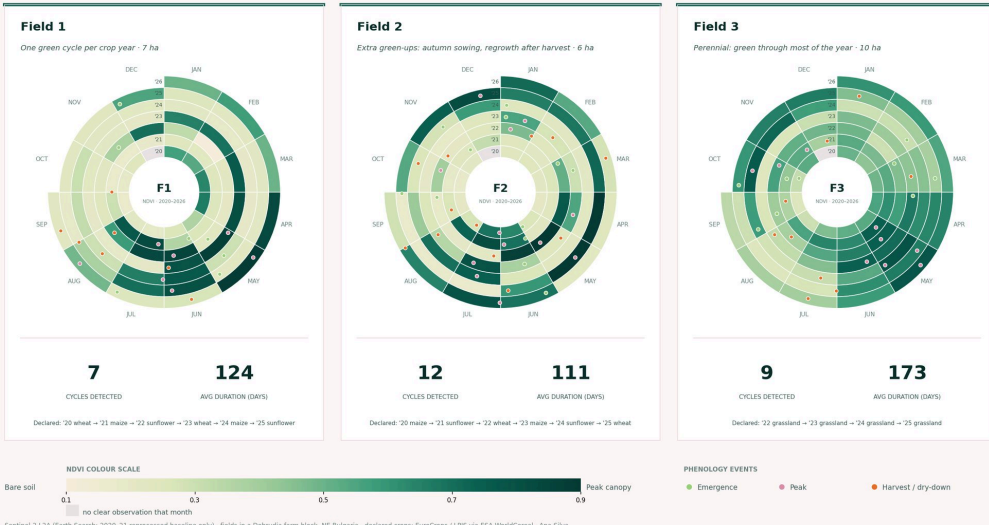
Selected work

Real Sentinel-1 and Sentinel-2 imagery · public parcel registers

Phenology clock: 2020-2026 at a glance

Each ring is one year · each wedge one month · colour = that month's median NDVI · dots mark detected emergence, peak and harvest/dry-down

JAN 2020 - SEP 202
SENTINEL-2 L2A - NOV
20% AMPLITUDE METHO

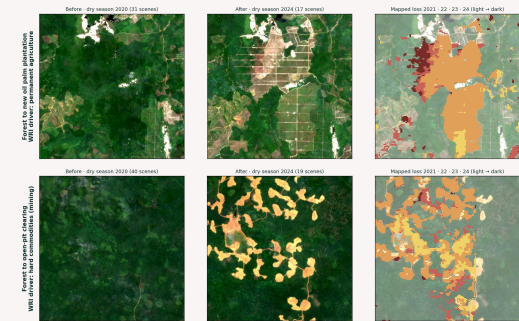


Phenology clock, seven years ring by ring

One ring per year on three neighbouring fields: rotations (wheat → maize → sunflower), emergence, peak and harvest, from 20,207 Sentinel-2 observations, 2020–2026.

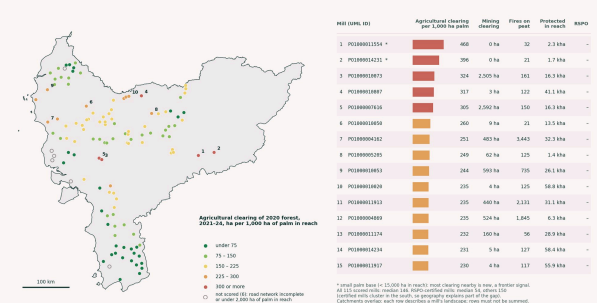
Not all clearing near palm oil mills is palm

Two 4 × 4 km windows in West Kalimantan, Sentinel-2 before (2020) and after (2024), with the mapped tree cover loss. Attributing each clearing to its driver keeps mining out of a palm deforestation score.



Which mills sit on an active agricultural clearing front

West Kalimantan: 2020 forest cleared 2021-24 where agriculture is the dominant driver, inside each mill's 2-hour sourcing landscape, per 1,000 ha of palm



Palm oil mills: where does the fruit come from?

361 mills in West Kalimantan and Riau, their truck-reachable palm, and forest cleared since the EUDR cutoff, split by cause.

Mills on the clearing front

Agricultural clearing of 2020 forest per 1,000 ha of palm in reach. A screening signal, shown by mill ID.

Fewer Dobrudzia fields left bare all summer since 2021

The drop came before the 2023 soil-cover rule, and the same fields show no clear change after it.

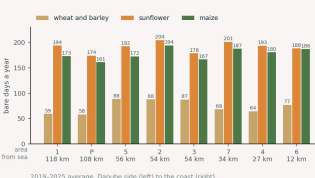
8 sample areas of 10 × 10 km, about 70,000 field-years, crop years 2019–2025, every clear Sentinel-2 date, weighted to the region's crop mix

The 8 sample areas, 10 × 10 km each (P = first pilot

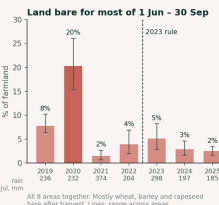


Sentinel-2, median of clear views 15 Jun – 20 Jul 2019

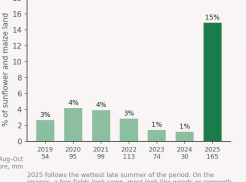
Bare days a year by crop, each sample area



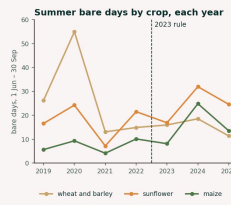
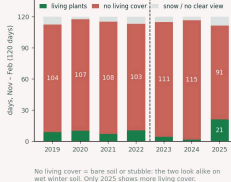
2010-2025 average. Dashed line (left) to the coast (right)



Sunflower and maize land green over winter



2025 follows the wettest late summer of the period. On the
 (image), a few fields look green, most look like woods or covered

er **Winter before sunflower and maize**

No living cover = bare soil or stubble: the two look alike on wet winter soil. Only 2025 shows more living cover.

Dobrudja: how long does the soil lie bare?

About 70,000 field-years in Bulgaria's grain belt, 2019–2025, every clear Sentinel-2 image. Fewer fields now sit bare all summer, but the drop came before the 2023 soil-cover rule, and winter is still the gap. Checked on the images.